Alex Ruvalc

Exam 1

1. Work in Python

a. $AB = \begin{bmatrix} 4 & -5 \\ 25 & 6 \end{bmatrix}$ AB is a 2x2 matrix.

b. $M = \left[\begin{array}{c|c} \begin{matrix} 1 & 2 \\ 0 & -1 \end{matrix} & I \\ \hline 0 & \begin{matrix} 1 & 2 \\ 0 & -1 \end{matrix} \end{array}\right]$

$M^2 = \begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$

c. $\|AB\|_F = 26.4953$

d. Induced 1-norm of C

$\|C\|_1 = 3$

e. $\{v_1, v_2, v_3\}$ is not linearly independent because the $\det(V) = 0$.

2. Distance through the Earth: 8674.7798 km
Distance along the surface of the Earth: 9543.2028 km

(calculations in Python)

Alex Buvolo

Exam 1

Work → 3a. We can conclude that the stacked vectors are linearly independent. If the a vectors are all linearly independent, then no matter if the b vectors are linearly independent or dependent, the c vectors will be linearly independent.

s in
rthan

b. We can not conclude that the stacked vectors are linearly dependent. If the a vectors are linearly dependent, then if the b vectors are linearly independent, the c vectors will also be linearly independent.

4. 400 stocks 250 trading days

$\underset{\text{rows}}{\underline{250}} \times \underset{\text{columns}}{\underline{400}}$ Max(rank) = 250

Rows can be lin. ind. not cols.

The supervisor is incorrect because it is guaranteed that at least 150 of the stocks are linearly dependent, since the max rank of a 250 × 400 matrix is 250. This has nothing to do with the companies being related. Most likely the intern did not make a mistake because she probably found that Google and some other stocks were dependent since it is certain that at least 150 are.

Alex Rinolo

Exam 1

5a. $b_t = \underbrace{(1+r)b_{t-1}}_{\text{prev bal + interest}} + \underbrace{c_t}_{\text{deposit/withdrawal}}$ $\quad r > 0 \quad \underbrace{t = 2, \dots, T}_{\text{periods}}$

r = interest rate

$b_1 = c_1$

Find $T \times T$ matrix A where $b = Ac$

$$c = \underbrace{\begin{bmatrix} 100 \\ -30 \\ 50 \\ 0 \\ -80 \end{bmatrix}}_{5\times 1} \qquad b = \underbrace{\begin{bmatrix} 100 \\ 80 \\ 138 \\ 151.8 \\ 86.98 \end{bmatrix}}_{5\times 1}$$

← Numbers from 5b (I did 5b first)

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1+r & 1 & 0 & 0 & 0 \\ (1+r)^2 & 1+r & 1 & 0 & 0 \\ (1+r)^3 & (1+r)^2 & 1+r & 1 & 0 \\ (1+r)^4 & (1+r)^3 & (1+r)^2 & 1+r & 1 \end{bmatrix}}_{5\times 5}$$

$$A = \underbrace{\begin{bmatrix} 1 & 0 & & & \cdots & 0 \\ (1+r) & 1 & & & & \\ (1+r)^2 & (1+r) & 1 & & & \\ (1+r)^3 & & & & & \\ \vdots & & & & & \\ & & & & & 0 \\ (1+r)^{T-1} & \cdots & (1+r)^3 & (1+r)^2 & (1+r) & 1 \end{bmatrix}}_{T\times T}$$

$b_2 \quad c_1 \cdot (1+r) + c_2$

$80 = 100 \cdot 1.1 - 30$

$b_3 \quad \overbrace{\qquad\qquad}^{b_2}$

$138 = (c_1(1+r) + c_2)(1+r) + c_3$

$138 = (c_1 + c_1 r + c_2)(1+r) + c_3$

$= c_1 + c_1 r + c_1 r + c_1 r^2 + c_2$

$+ c_2 r + c_3$

$= c_1 r^2 + 2c_1 r + c_1 + c_2 r$

$+ c_2 + c_3$

$= c_1(r^2 + 2r + 1) +$

$c_2(r+1) + c_3$

$r^2 + 2r + 1 = (r+1)^2$

5b. $b_t = (1+r)b_{t-1} + c_t$ $\quad t = 2, 3, 4, 5$

$T = 5 \quad r = 0.1$

$$c = \begin{bmatrix} 100 \\ -30 \\ 50 \\ 0 \\ -80 \end{bmatrix} \qquad \begin{matrix} c_1 = 100 \\ b_1 = 100 \end{matrix}$$

$b_2 = (1.1) \cdot 100 + (-30) = 80$

$b_3 = (1.1) \cdot 80 + 50 = 138$

$b_4 = (1.1) \cdot 138 + 0 = 151.8$

$b_5 = (1.1) \cdot 151.8 - 80 = 86.98$

$$b = \begin{bmatrix} 100 \\ 80 \\ 138 \\ 151.8 \\ 86.98 \end{bmatrix}$$

The bank account balance at the end is \$86.98.

Alex Ruvolo

Exam 1

$\alpha = \begin{bmatrix} 15 \\ -14 \\ -12 \\ -9 \\ -5 \end{bmatrix}$

(Bonus) 6. $c = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}_{5\times1}$ $r \geq 0$ let $r = 0$

1st row $c_1 \rightarrow$ 2nd row $c_2 \rightarrow$

$n = 5$ $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ \end{bmatrix}$ $\ell_1 = \begin{bmatrix} 1 \\ -(1+r) \\ 0 \\ \vdots \end{bmatrix}$ $\ell_2 = \begin{bmatrix} 0 \\ 1 \\ -(1+r) \\ 0 \end{bmatrix}$

$c = \alpha_1 e_1 + \alpha_2 \ell_1 + \alpha_3 \ell_2 + \alpha_4 \ell_3 + \alpha_5 \ell_4 + \alpha_n \ell_{n-1}$

$\alpha_1 = 1 + \frac{2}{1} + \frac{3}{1} + \frac{4}{1} + \frac{5}{1}$

$\alpha_1 = 15$

$d^T c = 1 + \frac{2}{1} + \frac{3}{1} + \frac{4}{1} + \frac{5}{1} = 15$

✓ $d^T c = \alpha_1$

$\overset{1}{c_1} = \alpha_1 \cdot 1 + \alpha_2 \cdot 1 + \alpha_3 \cdot 0 + \alpha_4 \cdot 0_2$ $\quad \overset{1}{c_1} = \overset{15}{\alpha_1} + \overset{-14}{\alpha_2}$

$\overset{2}{c_2} = \alpha_2 \cdot -(1+r) + \alpha_3 \cdot 1$ $\quad \overset{2}{c_2} = \overset{-12}{\alpha_3} - (1+r)\overset{-14}{\alpha_2}$

$\overset{3}{c_3} = \overset{-9}{\alpha_4} - (1+r)\overset{-12}{\alpha_3}$

Solve for: α_n

$\overset{4}{c_{n-1}} = \overset{-5}{\alpha_n} - (1+r)\overset{-9}{\alpha_{n-1}} \rightarrow$ ✓ $\overset{-5}{\alpha_n} = \overset{4}{c_{n-1}} + (1+r)\overset{-9}{\alpha_{n-1}}$

✓ $\overset{5}{c_n} = -(1+r)\overset{-5}{\alpha_n} \rightarrow$ ✓ $\overset{-5}{\alpha_n} = \frac{c_n}{-(1+r)}$ ✓ $-5 = \frac{5}{-1}$

Find $\alpha_{n-1} \rightarrow c_{n-1} - \alpha_n = -(1+r)\alpha_{n-1}$

✓ $\alpha_{n-1} = \frac{c_{n-1} - \alpha_n}{-(1+r)} \rightarrow -9 = \frac{4-(-5)}{-1}$ ✓ $-9 = -9$

The pattern continues for solving α, but I have no clue how it gets back to here.

Formula: $\alpha_1 = c_1 + \frac{c_2}{1+r} + \cdots + \frac{c_n}{(1+r)^{n-1}}$